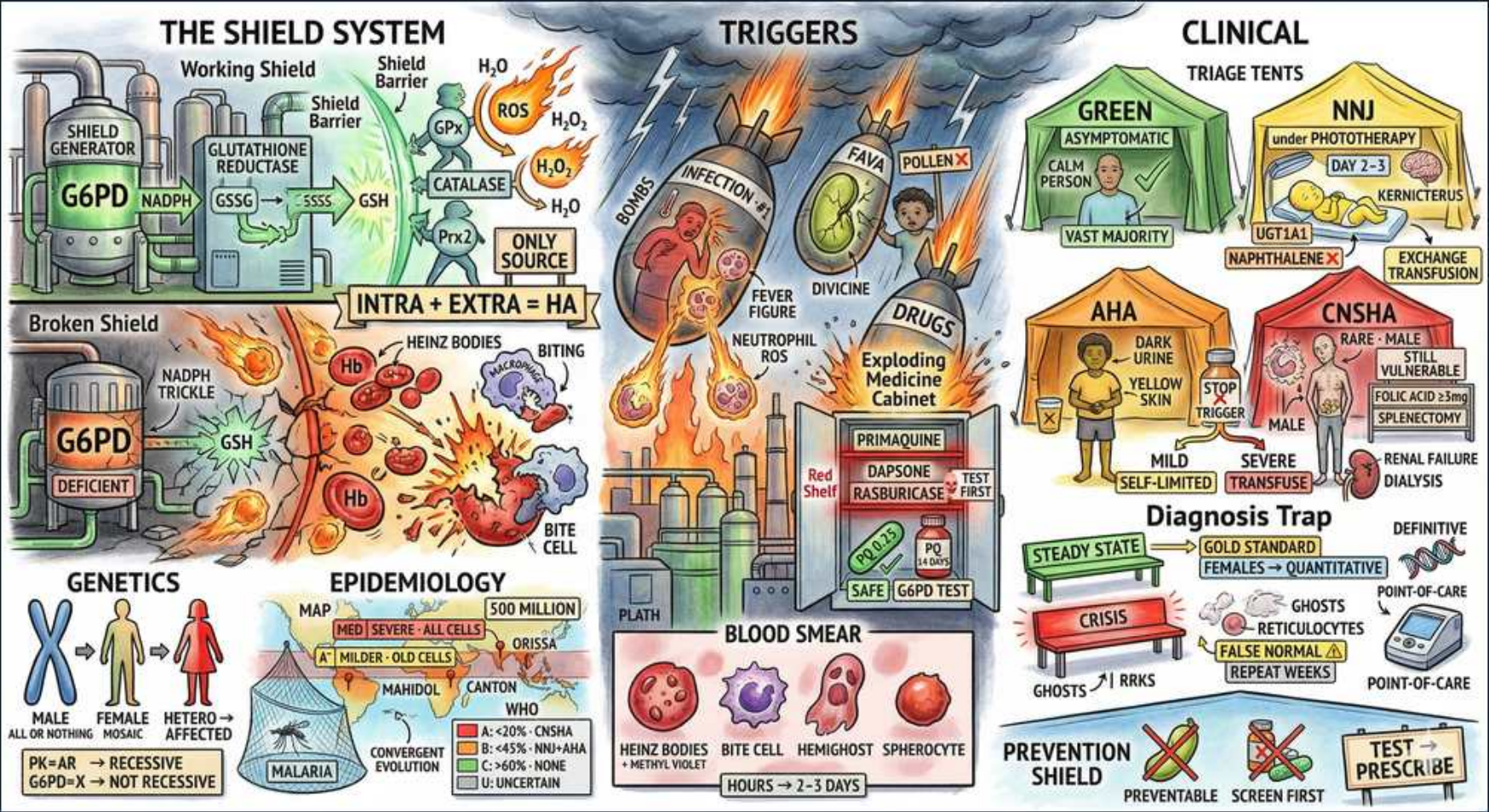


G6PD Deficiency: Oxidative Hemolysis



1. G6PD → NADPH shield

WHAT YOU SEE

Shield generator, NADPH

WHAT IT ENCODES

G6PD is the rate-limiting enzyme of the pentose phosphate pathway, generating NADPH. NADPH regenerates reduced glutathione (GSH), the RBC's only defense against oxidative stress. RBCs have no other NADPH source, so G6PD deficiency leaves them defenseless against oxidants.

BOARD TRAP

Thinking RBCs have backup NADPH sources; the HMP shunt via G6PD is the sole RBC NADPH supply.

2. Glutathione / catalase

WHAT YOU SEE

Glutathione reductase, catalase, ROS to H₂O

WHAT IT ENCODES

GSH and catalase neutralize H₂O₂/ROS. Without NADPH, GSH cannot be regenerated, oxidants accumulate, and hemoglobin is oxidized and precipitates. The pathway: G6PD → NADPH → GSH → detoxify peroxide. Loss of G6PD breaks this chain.

BOARD TRAP

Confusing the defect site; G6PD deficiency fails NADPH regeneration upstream, not catalase itself.

3. Heinz bodies

WHAT YOU SEE

Heinz body trickle, broken shield

WHAT IT ENCODES

Oxidized denatured hemoglobin precipitates as Heinz bodies (seen with crystal/methyl violet supravital stain). Splenic macrophages pluck them out, leaving membrane defects. Heinz bodies are the upstream lesion preceding bite cells.

BOARD TRAP

Looking for Heinz bodies on a routine Wright stain; they require a SUPRAVITAL stain (crystal violet), not standard smear.

4. Bite cells

WHAT YOU SEE

Biting RBC, bite cell

WHAT IT ENCODES

After the spleen removes Heinz bodies, the RBC has a scooped-out 'bite cell' (degmacyte). Bite cells and blister cells on smear during an oxidative crisis are the classic G6PD finding. They appear acutely with the trigger.

BOARD TRAP

Expecting bite cells on a baseline smear; they appear during acute oxidative hemolysis, not at steady state.

5. Triggers: infection

WHAT YOU SEE

Fire — infection, fever

WHAT IT ENCODES

INFECTION is the most common precipitant of G6PD hemolysis (oxidant burst from neutrophils). Other triggers: fava beans, drugs, and metabolic stress. Episodes are acute, self-limited as old cells are cleared and young cells take over.

BOARD TRAP

Naming drugs as the top trigger; infection is the single most common precipitant of acute G6PD hemolysis.

6. Fava beans

WHAT YOU SEE

Fava pod, pollen

WHAT IT ENCODES

Favism: ingestion of fava beans (vicine/convicine oxidants) triggers severe hemolysis, especially in the Mediterranean variant. Classic exam vignette is dark urine after a fava bean meal. Severity is variant-dependent.

BOARD TRAP

Expecting favism in all variants; severe favism is typical of the more severe Mediterranean variant, not mild A-.

7. Drug triggers

WHAT YOU SEE

Medicine cabinet — primaquine, dapson, rasburicase

WHAT IT ENCODES

Oxidant DRUGS: primaquine/tafenoquine, dapson, rasburicase, nitrofurantoin, sulfonamides, methylene blue. Rasburicase and methylene blue are absolutely contraindicated and can cause methemoglobinemia/hemolysis. Screen G6PD before rasburicase/primaquine.

BOARD TRAP

Giving rasburicase for tumor lysis or methylene blue for methemoglobinemia in a G6PD patient — both are contraindicated and worsen hemolysis.

8. X-linked recessive genetics

WHAT YOU SEE

Male all affected, female heterozygote

WHAT IT ENCODES

G6PD deficiency is X-LINKED RECESSIVE — males (hemizygous) are fully affected; heterozygous females are mosaic and usually mild. Contrast with PK deficiency, which is autosomal recessive. Predominantly affects males.

BOARD TRAP

Confusing inheritance with PK deficiency; G6PD is X-linked recessive, PK deficiency is autosomal recessive.

9. Malaria protection

WHAT YOU SEE

Epidemiology map, malaria, 500 million

WHAT IT ENCODES

G6PD deficiency is the most common human enzymopathy (~400-500 million), concentrated where malaria is endemic (Mediterranean, Africa, Asia) because it confers partial malaria resistance — convergent/balanced selection like sickle and thalassemia.

BOARD TRAP

Forgetting the malaria link; G6PD geographic distribution mirrors malaria endemicity due to selective advantage.

10. Blood smear timing

WHAT YOU SEE

Heinz bodies, bite cells, hemighost, spherocyte

WHAT IT ENCODES

During crisis: bite cells, blister cells, Heinz bodies (supravital), hemighosts. The picture evolves over 2-3 days. Self-limited because only the older, most enzyme-deficient cells lyse; reticulocytes are enzyme-replete.

BOARD TRAP

Expecting persistent hemolysis; G6PD crisis is self-limited as young high-enzyme cells replace lysed old cells.

11. Diagnosis timing trap

WHAT YOU SEE

Diagnosis trap — false-normal during crisis

WHAT IT ENCODES

Quantitative G6PD enzyme assay (gold standard) can be FALSELY NORMAL during/just after acute hemolysis because deficient old cells are gone and young reticulocytes have higher G6PD. Repeat testing 2-3 months later (steady state).

BOARD TRAP

Excluding G6PD deficiency on a normal assay drawn during the acute episode; retest at steady state weeks later.

12. Treatment / prevention

WHAT YOU SEE

Prevention shield, screen first, avoid triggers

WHAT IT ENCODES

Management is mainly avoidance of triggers (drugs, fava beans) and supportive care; severe cases need transfusion. Screen before high-risk drugs (rasburicase, primaquine, dapsone). Hydration protects kidneys during hemoglobinuria.

BOARD TRAP

Reaching for steroids or splenectomy; G6PD crisis is managed by removing the trigger and supportive care, not immunosuppression.

13. Green tents = mild majority

WHAT YOU SEE

Green triage, calm, vast majority

WHAT IT ENCODES

Class III (A-) variants cause episodic hemolysis only with triggers and are the vast majority — asymptomatic between episodes. Most G6PD-deficient people are well unless oxidatively stressed.

BOARD TRAP

Assuming all G6PD patients are chronically anemic; most are asymptomatic between trigger-induced episodes.

14. Neonatal jaundice (NNJ)

WHAT YOU SEE

Under phototherapy, kernicterus

WHAT IT ENCODES

G6PD deficiency is a major cause of neonatal jaundice and can lead to kernicterus; naphthalene (mothballs) is a classic neonatal trigger. Treat with phototherapy and exchange transfusion when severe. Screen at-risk newborns.

BOARD TRAP

Overlooking G6PD as a cause of severe neonatal jaundice; it is a leading cause requiring phototherapy/exchange.

15. CNSHA severe variant

WHAT YOU SEE

Class I, chronic, dialysis

WHAT IT ENCODES

Class I (CNSHA) variants cause chronic non-spherocytic hemolytic anemia even WITHOUT triggers — rare, severe, transfusion-dependent. Contrast with the common A-variant that hemolyzes only on exposure. Severity is variant-defined.

BOARD TRAP

Assuming G6PD only hemolyzes with triggers; severe Class I variants cause chronic ongoing hemolysis.
